

Experiment 03:

7. Laboratory Exercise

B. Program Code

1. HTML code:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Registration Form</title>
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
  </head>
  <body
    class="h-screen bg-gradient-to-b from-orange-300 via-orange-400 to-orange-500 flex items-
center justify-center p-6">
    <div class="w-full max-w-5xl bg-white rounded-3xl shadow-2xl p-8">
      <!-- Heading -->
      <div class="text-center mb-8">
        <h1 class="text-4xl font-bold text-gray-900">Registration Form</h1>
        <p class="mt-2 text-gray-500">
          Fill out the form carefully for registration
        </p>
      </div>
      <form class="space-y-8" id="form">
        <!-- Name -->
        <div>
          <h2 class="text-xl font-semibold mb-4">Student Name</h2>
          <div class="grid grid-cols-2 gap-6">
            <div>
              <input
                type="text"
                placeholder="First Name"
                id="fname"
                class="w-full rounded-lg border border-gray-300 px-4 py-3 focus:border-orange-
500 focus:ring-4 focus:ring-orange-200 outline-none"/>
            </div>
            <div>
              <input
                type="text"
```

```
        placeholder="Last Name"
        id="lname"
        class="w-full rounded-lg border border-gray-300 px-4 py-3 focus:border-orange-
500 focus:ring-4 focus:ring-orange-200 outline-none"/>
    </div>
</div>
</div>
<!-- Row 2 -->
<div class="grid grid-cols-2 gap-6">
    <div>
        <label class="block text-xl font-semibold mb-3"> Gender </label>
        <select
            id="gender"
            class="w-full rounded-lg border border-gray-300 px-4 py-3 focus:border-orange-500
focus:ring-4 focus:ring-orange-200 outline-none">
            <option value="" selected disabled>Please Select</option>
            <option value="Male">Male</option>
            <option value="Female">Female</option>
        </select>
    </div>
    <div>
        <label class="block text-xl font-semibold mb-3">
            Student E-mail
        </label>
        <input
            id="email"
            type="email"
            placeholder="myname@student.sfit.ac.in"
            class="w-full rounded-lg border border-gray-300 px-4 py-3 focus:border-orange-500
focus:ring-4 focus:ring-orange-200 outline-none"
        />
    </div>
</div>
<!-- Button -->
<div class="flex justify-center pt-2">
    <button
        type="submit"
        class="bg-emerald-500 hover:bg-emerald-600 text-white px-16 py-3 rounded-lg font-
medium shadow-lg transition hover:scale-105 hover:cursor-pointer">
        Submit
    </button>
</div>
```

```
        </form>
    </div>
</body>
<script src="./app.js"></script>
</html>
```

JS code:

```
const form = document.querySelector("#form");
const fname = document.querySelector("#fname");
const lname = document.querySelector("#lname");
const gender = document.querySelector("#gender");
const email = document.querySelector("#email");

form.addEventListener("submit", function (e) {
    e.preventDefault();

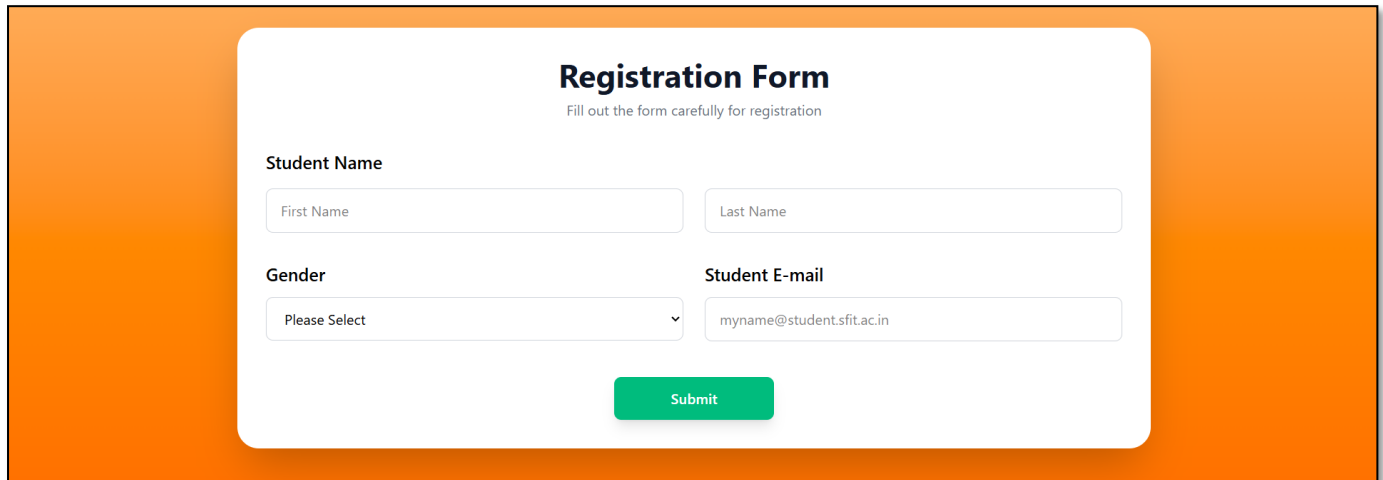
    // remove spaces
    let firstName = fname.value.trim();
    let lastName = lname.value.trim();
    let userEmail = email.value.trim();

    // First name validation
    if (firstName === "") {
        alert("Please enter your first name");
        return;
    }
    // Last name validation
    if (lastName === "") {
        alert("Please enter your last name");
        return;
    }
    // Gender validation
    if (gender.value === "") {
        alert("Please select your gender");
        return;
    }
    // Email validation
    if (userEmail === "") {
        alert("Please enter your email");
        return;
    }
    if (!validateEmail(userEmail)) {
        alert("Please enter a valid email address");
        return;
    }
    // Success
    alert("Registration successful!");
    form.reset();
});

// Email validation function
function validateEmail(email) {
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return emailPattern.test(email);
}
```

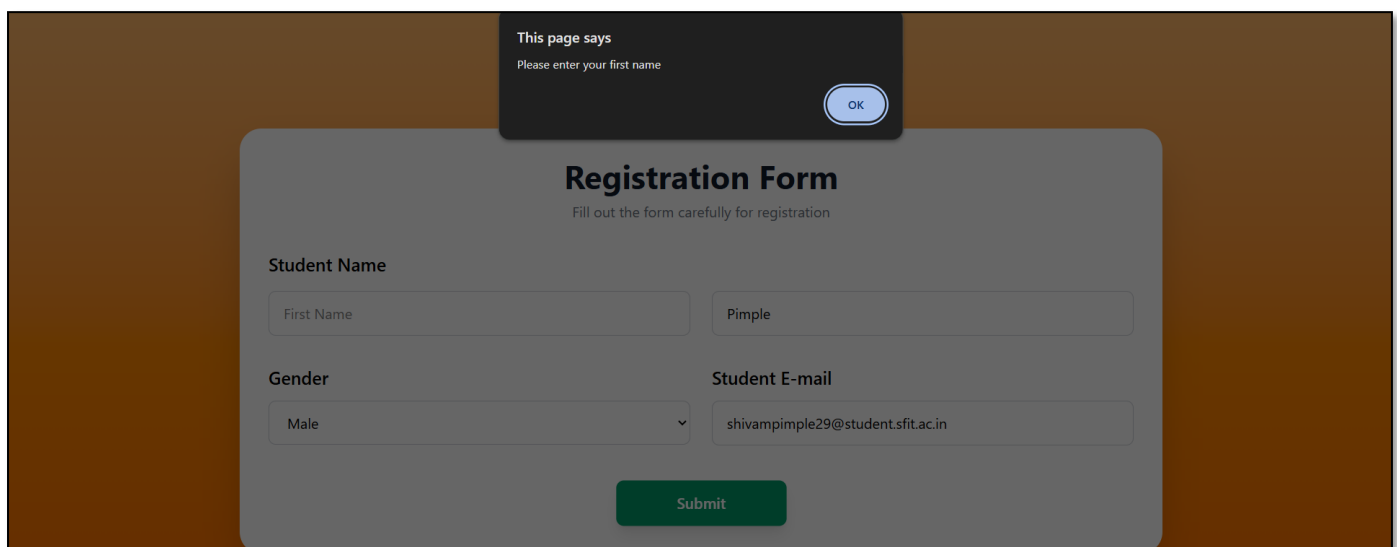
Output

1. Empty Form:



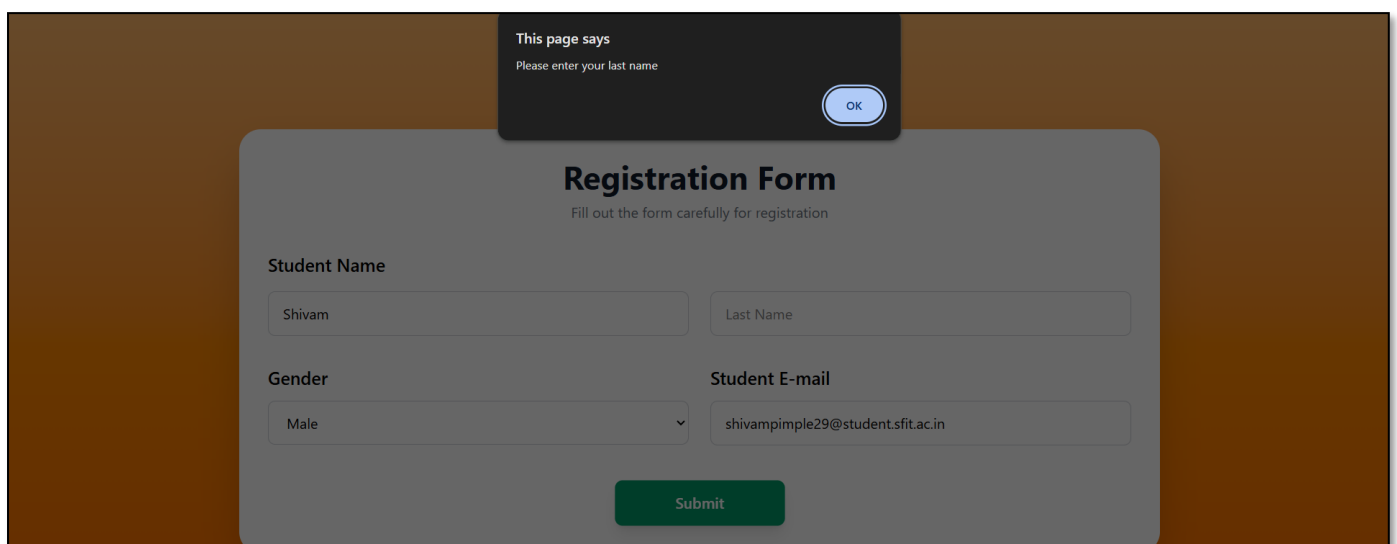
The screenshot shows a registration form titled "Registration Form" with the subtitle "Fill out the form carefully for registration". The form is set against an orange background. It contains four input fields: "First Name", "Last Name", "Gender" (a dropdown menu), and "Student E-mail". The "Student E-mail" field contains the text "myname@student.sfit.ac.in". A green "Submit" button is located at the bottom center of the form.

2. First name validation:



The screenshot shows the registration form with the "First Name" field filled with "Pimple". A dark grey validation message box appears at the top, stating "This page says Please enter your first name" with an "OK" button. The "Last Name" field is empty. The "Gender" dropdown is set to "Male". The "Student E-mail" field contains "shivampimple29@student.sfit.ac.in". The "Submit" button is green.

3. Last name validation:



The screenshot shows the registration form with the "Last Name" field filled with "Shivam". A dark grey validation message box appears at the top, stating "This page says Please enter your last name" with an "OK" button. The "First Name" field is empty. The "Gender" dropdown is set to "Male". The "Student E-mail" field contains "shivampimple29@student.sfit.ac.in". The "Submit" button is green.

4. Gender selection validation:

This page says
Please select your gender

OK

Registration Form

Fill out the form carefully for registration

Student Name

Shivam Pimple

Gender

Please Select

Student E-mail

shivampimple29@student.sfit.ac.in

Submit

5. Email validation:

This page says
Please enter your email

OK

Registration Form

Fill out the form carefully for registration

Student Name

Shivam Pimple

Gender

Male

Student E-mail

myname@student.sfit.ac.in

Submit

6. Successful Registration:

This page says
Registration successful!

OK

Registration Form

Fill out the form carefully for registration

Student Name

Shivam Pimple

Gender

Male

Student E-mail

shivampimple29@student.sfit.ac.in

Submit

Post Experimental Exercise:

1. Write JavaScript program to find factorial of a number.

Code:

```
let num = 5;
let factorial = 1;

if (num < 0) {
    console.log("Factorial does not exist for negative numbers.");
} else if (num === 0 || num === 1) {
    console.log("1");
} else {
    for (let i = 1; i <= num; i++) {
        factorial *= i;
    }
    console.log("Factorial of " + num + " is " + factorial);
}
```

Output:

```
$ node index.js
Factorial of 5 is 120
```

2. Write JavaScript program to print Fibonacci series up to 10 numbers.

Code:

```
let a = 0;
let b = 1;

console.log("Fibonacci Series:");

for (let i = 1; i <= 8; i++) {
    console.log(a);

    let c = a + b;
    a = b;
    b = c;
}
```

Output:

```
Admin@DESKTOP-P0P8PIU MINGW64 /d/WebX_exp3
$ node index.js
0
1
1
2
3
5
8
13
```